LSM6DS3 programming

Document	What it says about 4th data set
LSM6DS3	"4th data set = step counter and timestamp info" — no byte map
AN4650 (Sec 8.9)	Full byte-level layout of the 4th data set
(lsm6ds3.c)	Working C code that unpacks it

Read write protocol 6.2 (SPI)
The LSM6DS3 is a bus slave, the SPI allows writing and reading the registers of the device. CS is the serial port in able and it is controlled by the SPI master. It goes low at the start of the transmission and goes back high at the end. The transmission is completed in a 16 clock cycles where:

Write:
Bit 0: R/W , when 0 write.
Bit 1-7: Intended address, by using c operator AND with 0x7, (0x7F & 0XFF) = 0x7F bits 1-7 can be transmitted as register values while still using bit 0 as R/W indicator.
Bit 8-15: used for writing the data being transmitted.

Read:
Bit0: R/W bit. The value is 1 when reading
Bit1-7: address of indexed register, by using C operator OR with 0x80, (0x80 | 0x00) = 0x80, only bit 0 turns back to 1 keeping bits 1-7 for reg value.
Bit 8-15: Used for data being received.

Registers used Sec 8, Table 16

Register Name	Address
WHO_AM_I	0x0F
FIFO_CTRL1	0x06
FIFO_CTRL2	0x07
FIFO_CTRL3	0x08
FIFO_CTRL4	0x09
FIFO_CTRL5	0x0A
CTRL1_XL	0x10
CTRL2_G	0x11
CTRL3_C	0x12
TAP_CFG	0x58
WAKE_UP_DUR	0x5C
FIFO_STATUS1	0x3A
FIFO_STATUS2	0x3B
FIFO_DATA_OUT_L	0x3E
TIMESTAMP0_REG	0x40
TIMESTAMP1_REG	0x41
TIMESTAMP2_REG	0x42

FIFO Sec 5.4.6
Table 143-146] FIFO_DATA_OUT_L / FIFO_DATA_OUT_H
The FIFO on the IMU consist of sending 6 16 bit bytes or 12 8 bit bytes portioned as Low & High bytes, the order of the data is broken as (gyro_x_l [0], gyro_x_h[1], gyro_y_l[2], gyro_y_h[3], gyro_z_l[4], gyro_z_h[5]) & from 6-11 for Acc.

Time stamp 9.58
Time stamp comes as a 24bit word, or 3 bytes. Each value can be read in 24 clock cycles, starting from Timestamp0 it can be buffered into a uint32 using a 3 byte array.

IMU init:

Accelerometer config CTRL1_XL

Field	Description
ODR_XL [3:0]	Output data rate and power mode selection. Default value: 0000 (Table 47).
FS_XL [1:0]	Accelerometer full-scale selection. Default value: 00 (00: ±2 g; 01: ±16 g; 10: ±4 g; 11: ±8 g)
BW_XL [1:0]	Anti-aliasing filter bandwidth selection. Default value: 00 (00: 400 Hz; 01: 200 Hz; 10: 100 Hz; 11: 50 Hz)

ODR_XL3	ODR_XL2	ODR_XL1	ODR_XL0	ODR [Hz] when XL_HM_MODE = 1	ODR [Hz] when XL_HM_MODE = 0
0	0	0	0	Power-down	Power-down
0	0	0	1	12.5 Hz (low power)	12.5 Hz (high performance)
0	0	1	0	26 Hz (low power)	26 Hz (high performance)
0	0	1	1	52 Hz (low power)	52 Hz (high performance)
0	1	0	0	104 Hz (normal mode)	104 Hz (high performance)
0	1	0	1	208 Hz (normal mode)	208 Hz (high performance)
0	1	1	0	416 Hz (high performance)	416 Hz (high performance)
0	1	1	1	833 Hz (high performance)	833 Hz (high performance)
1	0	0	0	1.66 kHz (high performance)	1.66 kHz (high performance)
1	0	0	1	3.33 kHz (high performance)	3.33 kHz (high performance)
1	0	1	0	6.66 kHz (high performance)	6.66 kHz (high performance)

ODR	XL_BW_SCAL_ODR = 0	XL_BW_SCAL_ODR = 1
6.66 – 3.33 kHz	Filter not used	—
1.66 kHz	400 Hz	Bandwidth is determined by setting BW_XL[1:0] in CTRL1_XL (10h)
833 Hz	400 Hz	"
416 Hz	200 Hz	"
208 Hz	100 Hz	"
104 – 12.5 Hz	50 Hz	"

ODR_XL3	ODR_XL2	ODR_XL1	ODR_XL0	FS_XL1	FS_XL0	BW_XL1	BW_XL0
1	0	0	0	0	1	0	0

Config (reg | hex value): (0x10 |0x84)
this configuration is a selected due to high performance and high G maneuvers, will be changed to 8g since its more practical for airplanes and navigation. 16g is mad (will go over tis again).

CTRL2_G register description

Field	Description
ODR_G [3:0]	Gyroscope output data rate selection. Default value: 0000 (Refer to Table 51).
FS_G [1:0]	Gyroscope full-scale selection. Default value: 00 (00: 250 dps; 01: 500 dps; 10: 1000 dps; 11: 2000 dps)
FS_125	Gyroscope full-scale at 125 dps. Default value: 0 (0: disabled; 1: enabled)

ODR_G3	ODR_G2	ODR_G1	ODR_G0	ODR [Hz] when G_HM_MODE = 1	ODR [Hz] when G_HM_MODE = 0
0	0	0	0	Power down	Power down
0	0	0	1	12.5 Hz (low power)	12.5 Hz (high performance)
0	0	1	0	26 Hz (low power)	26 Hz (high performance)
0	0	1	1	52 Hz (low power)	52 Hz (high performance)
0	1	0	0	104 Hz (normal mode)	104 Hz (high performance)
0	1	0	1	208 Hz (normal mode)	208 Hz (high performance)
0	1	1	0	416 Hz (high performance)	416 Hz (high performance)
0	1	1	1	833 Hz (high performance)	833 Hz (high performance)
1	0	0	0	1.66 kHz (high performance)	1.66 kHz (high performance)

ODR_G3	ODR_G2	ODR_G1	ODR_G0	FS_G1	FS_G0	FS_125	0^(1)
1	0	0	0	1	1	0	0

R/W config CTRL3_C

Bit Field	Description
BOOT	Reboot memory content. Default value: 0 (0: normal mode; 1: reboot memory content[1])
BDU	Block Data Update. Default value: 0 (0: continuous update; 1: output registers not updated until MSB and LSB have been read)
H_LACTIVE	Interrupt activation level. Default value: 0 (0: interrupt output pads active high; 1: interrupt output pads active low)
PP_OD	Push-pull/open-drain selection on INT1 and INT2 pads. Default value: 0 (0: push-pull mode; 1: open-drain mode)
SIM	SPI Serial Interface Mode selection. Default value: 0 (0: 4-wire interface; 1: 3-wire interface)
IF_INC	Register address automatically incremented during a multiple byte access with a serial interface (I^2C or SPI). Default value: 1 (0: disabled; 1: enabled)
BLE	Big/Little Endian Data selection. Default value: 0 (0: data LSB @ lower address; 1: data MSB @ lower address)
SW_RESET	Software reset. Default value: 0 (0: normal mode; 1: reset device) This bit is cleared by hardware after next flash boot.

Boot [7]	BDU [6]	H_LACTIVE[5]	PP_OD[4]	SIM[3]	IF_IN[2]	BLE[1]	SW_RESET[0]
0	1	0	0	0	1	0	0

Config (reg | hex value): (0x12 |0x44)
This register defines the interfacing to the device since by setting BDU high only allows data to be uploaded once new data has arrives for acc and gyro, this gives it a more portioned set of data with no repeated measurements.

TAP_CFG timestamp

Bit Field	Description
TIMER_EN	Timestamp count enable, output data are collected in TIMESTAMP0_REG (40h), TIMESTAMP1_REG (41h), TIMESTAMP2_REG (42h) register. Default: 0 (0: timestamp count disabled; 1: timestamp count enabled)
PEDO_EN	Pedometer algorithm enable. Default value: 0 (0: pedometer algorithm disabled; 1: pedometer algorithm enabled)
TILT_EN	Tilt calculation enable. Default value: 0 (0: tilt calculation disabled; 1: tilt calculation enabled)
SLOPE_FDS	Enable accelerometer HP and LPF2 filters (refer to Figure 6). Default value: 0 (0: disable; 1: enable)
TAP_X_EN	Enable X direction in tap recognition. Default value: 0 (0: X direction disabled; 1: X direction enabled)
TAP_Y_EN	Enable Y direction in tap recognition. Default value: 0 (0: Y direction disabled; 1: Y direction enabled)
TAP_Z_EN	Enable Z direction in tap recognition. Default value: 0 (0: Z direction disabled; 1: Z direction enabled)
LIR	Latched Interrupt. Default value: 0 (0: interrupt request not latched; 1: interrupt request latched)

TIMER_EN	PEDO_EN	TILT_EN	SLOPE_FDS	TAP_X_EN	TAP_Y_EN	TAP_Z_EN	LIR
1	0	0	0	0	0	0	0

WAKE_UP_DUR

Bit Field	Description
FF_DUR5	Free fall duration event. Default: 0 For the complete configuration of the free-fall duration, refer to FF_DUR[4:0] in FREE_FALL (5Dh) configuration.
WAKE_DUR [1:0]	Wake up duration event. Default: 00 1 LSB = 1 ODR_time
TIMER_HR	Timestamp register resolution setting[1]. Default value: 0 (0: 1 LSB = 6.4 ms; 1: 1 LSB = 25 μs)
SLEEP_DUR [3:0]	Duration to go in sleep mode. Default value: 0000 1 LSB = 512 ODR

FF_DUR	WAKE_DUR1	WAKE_DUR0	TIMER_HR	SLEEP_DUR3	SLEEP_DUR2	SLEEP_DUR1	SLEEP_DUR0
0	0	0	1	0	0	0	0

Config (reg | hex value): (0x5C |0x10)
This config sets up the timestamp resolution, using 25us allows for more accurate integration between raw sensor reads.

FIFO CTRL1

BIT FIELDS	REG DESCRIPTION
FTH_[7:0]	FIFO threshold level setting(1). Default value: 0000 0000. Watermark flag rises when the number of bytes written to FIFO after the next write is greater than or equal to the threshold level. Minimum resolution for the FIFO is 1 LSB = 2 bytes (1 word) in FIFO

FTH_7	FTH_6	FTH_5	FTH_4	FTH_3	FTH_2	FTH_1	FTH_0
0	0	0	0	1	0	0	1

e

Config (reg | hex value): (0x06 | 0x09)

This register takes a value and flags FIFO is full, threshold level which creates a watermark signaling that FIFO data is available. Value 0x9 signifies that there's a total of 9 16 bit bytes.

FIFO CTRL2

Bit Field	Description
TIMER_PEDO_FIFO_EN	Enable pedometer step counter and timestamp as 4th FIFO data set. Default: 0 (0: disable step counter and timestamp data as 4th FIFO data set; 1: enable step counter and timestamp data as 4th FIFO data set)
TIMER_PEDO_FIFO_DRDY	FIFO write mode[1]. Default: 0 (0: enable write in FIFO based on XL/Gyro data-ready; 1: enable write in FIFO at every step detected by step counter)
FTH_[11:8]	FIFO threshold level setting[2]. Default value: 0000 Watermark flag rises when the number of bytes written to FIFO after the next write is greater than or equal to the threshold level. Minimum resolution for the FIFO is 1 LSB = 2 bytes (1 word) in FIFO

FEFO_PERDO_ FIFO_EN	TIMER_PEDO_FIF O_DRDY	0^1	0^1	FTH_11	FTH_10	FTH_9	FTH_8
1	0	0	0	0	0	0	0

Config (reg | hex value): (0x07 | 0x80)

This register enables the pedometer step counter and timestamp as 4th FIFO Data set.

FIFO CTRL3

Bit Field	Description
DEC_FIFO_GYRO [2:0]	Gyro FIFO (first data set) decimation setting. Default: 000 For the configuration setting, refer to Table 27.
DEC_FIFO_XL [2:0]	Accelerometer FIFO (second data set) decimation setting. Default: 000 For the configuration setting, refer to Table 28.

DEC_FIFO_GYRO [2:0]	Configuration
000	Gyro sensor not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

DEC_FIFO_XL [2:0]	Configuration
000	Accelerometer sensor not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

0^ 1	0^ 1	DEC_FIFO_GYRO 2	DEC_FIFO_GYRO 1	DEC_FIFO_GYRO 0	DEC_FIFO_XL 0	DEC_FIFO_XL 1	DEC_FIFO_XL 0
0	0	0	0	1	0	0	1

Config (reg | hex value): (0x08 | 0x09)

This register allows to add decimation to the raw values, no decimation selected.

FIFO_CTRL4

Bit Field	Description
ONLY_HIGH_DATA	8-bit data storage in FIFO. Default: 0 (0: disable MSByte only memorization in FIFO for XL and Gyro; 1: enable MSByte only memorization in FIFO for XL and Gyro in FIFO)
DEC_DS4_FIFO [2:0]	Fourth FIFO data set decimation setting. Default: 000 For the configuration setting, refer to Table 31.
DEC_DS3_FIFO [2:0]	Third FIFO data set decimation setting. Default: 000 For the configuration setting, refer to Table 32.

DEC_DS4_FIFO [2:0]	Configuration
000	Fourth FIFO data set not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

DEC_DS3_FIFO [2:0]	Configuration
000	Third FIFO data set not in FIFO
001	No decimation
010	Decimation with factor 2
011	Decimation with factor 3
100	Decimation with factor 4
101	Decimation with factor 8
110	Decimation with factor 16
111	Decimation with factor 32

0^1	ONLY_HIGH_DATA	DEC_DS4_FIFO2	DEC_DS4_FIFO1	DEC_DS4_FIFO0	DEC_DS3_FIFO2	DEC_DS3_FIFO1	DEC_DS3_FIFO0
0	0	0	0	1	0	0	0

Config (reg | hex value): (0x09 | 0x08)

This register controls the decimation for the 4th dataset from FIFO which is used for timestamp In this case.

FIFO_CTRL5

Bit Field	Description
ODR_FIFO_ [3:0]	FIFO ODR selection, setting FIFO_MODE also. Default: 0000 For the configuration setting, refer to Table 35.
FIFO_MODE_ [2:0]	FIFO mode selection bits, setting ODR_FIFO also. Default value: 000 For the configuration setting, refer to Table 36.

ODR_FIFO_ [3:0]	Configuration[1]
0000	FIFO disabled
0001	FIFO ODR is set to 12.5 Hz
0010	FIFO ODR is set to 26 Hz
0011	FIFO ODR is set to 52 Hz
0100	FIFO ODR is set to 104 Hz
0101	FIFO ODR is set to 208 Hz
0110	FIFO ODR is set to 416 Hz
0111	FIFO ODR is set to 833 Hz
1000	FIFO ODR is set to 1.66 kHz
1001	FIFO ODR is set to 3.33 kHz
1010	FIFO ODR is set to 6.66 kHz

FIFO_MODE_ [2:0]	Configuration mode
000	Bypass mode. FIFO disabled.
001	FIFO mode. Stops collecting data when FIFO is full.
010	Reserved
011	Continuous mode until trigger is deasserted, then FIFO mode.
100	Bypass mode until trigger is deasserted, then Continuous mode.
101	Reserved
110	Continuous mode. If the FIFO is full, the new sample overwrites the older one.
111	Reserved

0^1	ODR_FIFO_3	ODR_FIFO_2	ODR_FIFO_1	ODR_FIFO_0	FIFO_MODE2	FIFO_MODE1	FIFO_MODE0
0	1	0	0	0	0	0	0

0^1	ODR_FIFO_3	ODR_FIFO_2	ODR_FIFO_1	ODR_FIFO_0	FIFO_MODE2	FIFO_MODE1	FIFO_MODE0
0	1	0	0	0	1	1	0

1 Config (reg | hex value): (0x0A | 0x08)
Initially after config set disable FIFO bypass mode to flush any data in the buffer

2 Config (reg | hex value): (0x0A | 0x08)
Afterwards configure FIFO for Continuous mode. If the FIFO is full, the new sample overwrites the older one.